



Spring 2024

Electronics for Instrumentation

Lab. 2: Introduction to LabVIEW

Lab Objectives ^a

- 1- Identify VI parts and navigate palettes.
- 2- Use LabVIEW Help.
- 3- Create VI.
- 4- Create State Machine

1. Exercise 1: Exploring a VI (Identify the parts of an existing VI)

- **Description:** This VI takes the seconds until a plane arrives at an airport and converts the time into a combination of hours/minutes/seconds. This VI can display the remaining time until the plane arrives.

- **Steps:**

1. On the front panel shown in Figure 1.1, identify the following items: *Control, Indicator, Run button, and Icon*

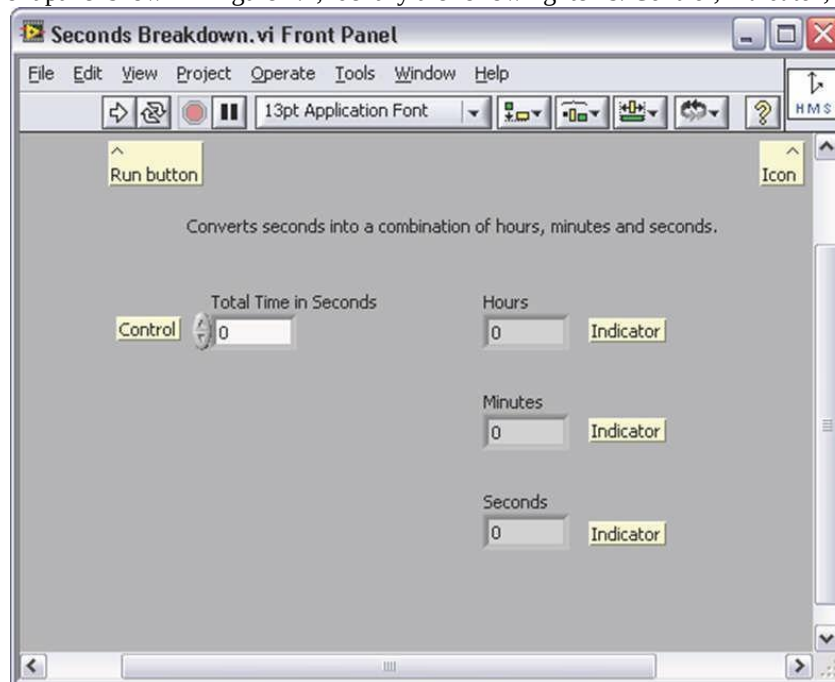


Figure 1.1 Front Panel Items

2. On the block diagram shown in Figure 1.2, identify the following items: *Control, Indicator, Constant, and Free Label*
3. Test manually the VI (block diagram in Figure 1.2) using the values given in Table 1.1.
 - Enter the input value in the **Total Time in Seconds** control.
 - Imagine you clicked the **Run** button.
 - For each input, calculated the given outputs and then compare them to the outputs listed in Table 1.1.

Table 1.1 Testing Values for Seconds Breakdown.vi

Input	Output
0 seconds	0 hours, 0 minutes, 0 seconds
60 seconds	0 hours, 1 minute, 0 seconds
3600 seconds	1 hour, 0 minutes, 0 seconds
3665 seconds	1 hour, 1 minute, 5 seconds

^a This Lab Sheet is based on NI training material

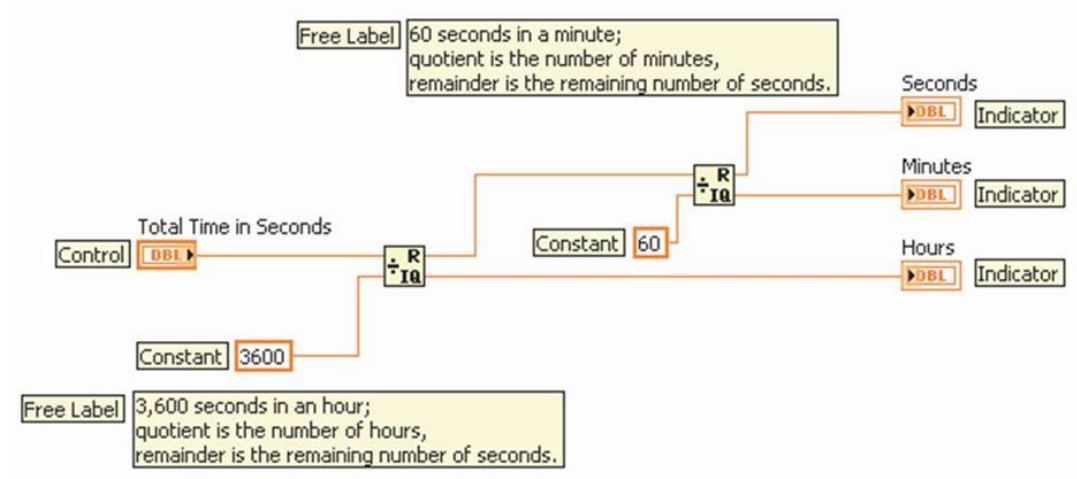


Figure 1.2 Block Diagram Items

2. Exercise 2: Navigating Palettes (Learn to find controls and functions)

Steps:

1. **Open** a blank VI
2. Select **View»Controls Palette** on the front panel window.
3. Explore the Controls palette.
 - ❖ Click the **Search** button.
 - ❖ Type *string control*.
 - ❖ Click a search result and drag it to the front panel window to place the object.
4. To view the front panel and block diagram at the same time, select **Window»Tile Up and Down**.
5. Open the block diagram and select **View»Functions Palette**.
4. Explore the **Functions** palette.
 - ❖ Place the “Wait (ms)” block in the **Favorites** Category.
 - Locate the “Wait (ms)” block.
 - On the **Timing** subpalette, right-click on the “Wait (ms)” block and select **Add Item to Favorites** from the shortcut menu.
 - Notice that the **Favorites** category on the **Functions** palette now contains the “Wait (ms)” block.
 - ❖ Practice accessing similar functions.
 - Place an Add function on the block diagram.
 - Right-click the Add function and notice that a **Numeric** palette is available. ➤ Practice placing functions from the **Numeric** palette on the block diagram.

3. Exercise 3: Using Help (Become familiar with using the Context Help window, the LabVIEW Help, and the NI Example Finder).

- **Description:** This exercise consists of a series of tasks designed to help you become familiar with the LabVIEW Help tools.

NI Example Finder

1. Imagine that you want to learn how to concatenate a string in a LabVIEW program that you are developing. Use the NI Example Finder to find a VI that concatenates two strings.
 - ❖ Open LabVIEW.
 - ❖ Select **Help»Find Examples** to open the NI Example Finder.
 - ❖ Confirm that the **Task** option is selected on the **Browse** tab.
 - ❖ Double-click the **Fundamentals** folder.
 - ❖ Select the **Strings** sub-folder.
 - ❖ Select **Concatenate Strings.vi**. Notice that a description of the VI is provided in the **Information** text box so that you can verify that this VI meets your needs.
 - ❖ Double-click **Concatenate Strings.vi** to open the VI.
 - ❖ Explore this VI and its components.
 - ❖ Close the VI after you finish exploring it.

2. You want to learn more about using Express VIs to filter signals. Use the NI Example Finder to find an appropriate VI.
 - ❖ The NI Example Finder should still be open from the previous step. If not, open the NI Example Finder.
 - ❖ Click the **Search** tab in the NI Example Finder.
 - ❖ Enter express in the **Enter keyword(s)** field to find VIs that contain Express VIs.
 - ❖ Double-click the Express result that appears in the **Double-click keyword(s)** field.
 - ❖ This keyword is associated with many example VIs, as demonstrated by the number of VIs returned. You can select any one of these VIs and read the description in the **Information** text box.
 - ❖ Double-click **Express Filter.vi** to open it.
 - **Context Help Window**
 3. Use the **Context Help** window to learn about the Express VIs used in the Express Filter VI.
 - ❖ Open the block diagram of the **Express Filter.vi** by selecting **Window»Show Block Diagram**.
 - ❖ Open the **Context Help** window by selecting **Help»Show Context Help**.
 - ❖ Move the **Context Help** window to a convenient area where the window does not hide part of the block diagram.
 - ❖ Place your mouse cursor over the Simulate Signal Express VI. The **Context Help** window content changes to show information about the object that your mouse is over.
 - ❖ Move your mouse over another Express VI. Notice the **Context Help** window content changes corresponding to the location of the mouse cursor.
 - ❖ Move your mouse over one of the Filter Express VIs.
 - ❖ Examine the configuration details in the **Context Help** window. This gives you the information about how the Express VI is configured.
 - ❖ Double-click the Filter Express VI to open the configuration dialog box. Notice that the selections in the configuration dialog box match the information in the **Context Help** window.
 - ❖ Click the **OK** button to close the configuration dialog box.
 4. Anchor the **Context Help** window so that you can move your mouse without the contents of the window changing. The **Context Help** window should show information about the Simulate Signal Express VI.
 - ❖ Move your mouse over the Simulate Signal Express VI.
 - ❖ To anchor the context help window, select the **Lock** button in the lower left corner of the window.
- Tip** If the contents of the window change before you lock the window, avoid passing your mouse over other objects on the way to the **Context Help** window. Move the window closer to the object of interest to view **Context Help** for that item.
- ❖ Move your mouse over another object. Notice the contents of the window do not change while the **Lock** button (in the bottom of the context help window) is selected.
 - ❖ Deselect the **Lock** button to resume normal operation of the window.
5. Modify the **Description and Tip** associated with the **Simulated frequency** control to change the content shown in the **Context Help** window.
 - ❖ Select **Window»Show Front Panel** to open the front panel of the VI.
 - ❖ Move your mouse over the **Simulated frequency** control.
 - ❖ Read the contents of the **Context Help** window.
 - ❖ Right-click the **Simulated frequency** control.
 - ❖ Select **Description and Tip** from the shortcut menu.
 - ❖ Replace the text in the "**Simulated frequency**" **Description** box with the text: This is the description of the control.
 - ❖ Replace the text in the "**Simulated frequency**" **Tip** box with the text: This is the tip for the control.
 - ❖ Click the **OK** button.
 - ❖ Move your mouse over the **Simulated frequency** control.

- ❖ Notice that the contents of the **Context Help** window changed to match the text you typed in the **Description** field of the **Description and Tip** dialog box.
- ❖ Run the VI.
- ❖ Place your mouse cursor over the **Simulated frequency** control.
- ❖ Notice that the tool tip that appears matches the text you typed in the **Tip** field of the **Description and Tip** dialog box.
- ❖ Click the **Stop** button.

▪ **LabVIEW Help**

6. Use the *LabVIEW Help* to learn more information about the Filter Express VI.
 - ❖ Select **Window»Show Block Diagram** to open the block diagram of the Express Filter VI.
 - ❖ Right-click the Filter Express VI and select **Help** from the shortcut menu. This opens the *LabVIEW Help* topic for the Filter Express VI.

Note To access the *LabVIEW Help* for this topic, you can also select the Detailed Help link in the **Context Help** window while the Filter Express VI is selected, or click the question mark in the **Context Help** window.

- ❖ Explore the topic. For example, what is the purpose of the **Cutoff Frequency (Hz)** dialog box option?
- ❖ Close the *LabVIEW Help* window.

7. Close the Express Filter VI when you finish. Do not save changes.

4. Exercise 4: Determine Warnings VI (Create and document a simple VI.)

- **Scenario:** You must create a portion of a larger project. The lead developer gives you the inputs of the VI, the algorithm, and the expected outputs. Create and document a VI based on the given design.

- **Design:** Inputs and Outputs

Type	Name	Properties
Numeric control	Current Angle	Double-precision, floating-point
Numeric control	Max Angle	Double-precision, floating-point
Numeric control	Min Angle	Double-precision, floating-point
String indicator	Warning Text	Three potential values: Upper Limit Warning, No Warning, and Lower Limit Warning
Round LED indicator	Warning?	

- **Flowchart:** see Figure 4.1
- **Implementation:** Follow the instructions given below to create a front panel similar to Figure 4.2. The user enters the current angle, maximum angle, and minimum angle. Then, the front panel displays the warning string and the warning Boolean LED.
 1. Open a blank VI and create the following front panel (shown in Figure 4.2).
 2. Save the new VI.
 - ❖ Select **File»Save**.
 - ❖ Save the VI as Determine Warnings.vi
 3. Create a numeric control for the current angle.
 - ❖ Add a numeric control to the front panel window.
 - ❖ Change the label of the numeric control to Current Angle.
 - ❖ Right-click the control, select **Representation**, and confirm that the representation type is set to double precision.



Tip This subVI could be also be used for Radians, as long as all inputs use the same scale. Therefore, it is not necessary to add scale units to the labels.

4. Create a numeric control for the maximum angle.
 - ❖ Hold down the <Ctrl> key and click and drag the Current Temp numeric control to create a copy of the control.
 - ❖ Change the label text of the new numeric control to Max Angle.

5. Create a numeric control for the minimum angle.
 - ❖ Hold down the <Ctrl> key and click and drag the Max Temp numeric control to create a copy of the control.
 - ❖ Change the label text of the new numeric control to Min Angle.
6. Create a string indicator for the warning text.
 - ❖ Add a string indicator to the front panel window.
 - ❖ Change the label text of the string indicator to Warning Text.
7. Create a Round LED or other Boolean indicator for the warning Boolean.
 - ❖ Add a Round LED to the front panel window.
 - ❖ Change the label text of the Boolean indicator to Warning?.
8. Switch to the block diagram.

Tip If you do not want to view terminals as icons on the block diagram, select **Tools» Options**, then select **Block Diagram** from the **Category** list. Remove the checkmark from the **Place front panel terminals as icons** item.

9. Complete the following instructions to create a block diagram similar to Figure 4.3.

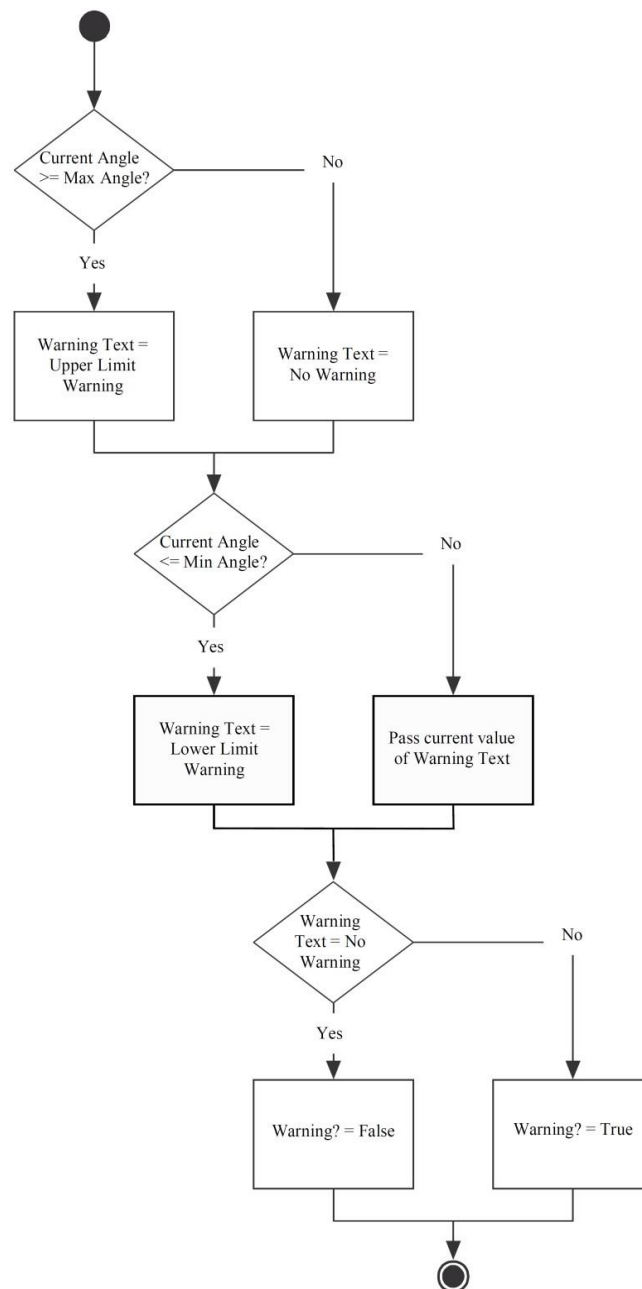


Figure 4.1 Determine Warnings VI Flowchart

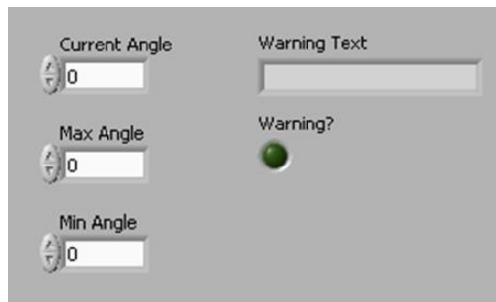


Figure 4.2 Determine Warnings VI Front Panel

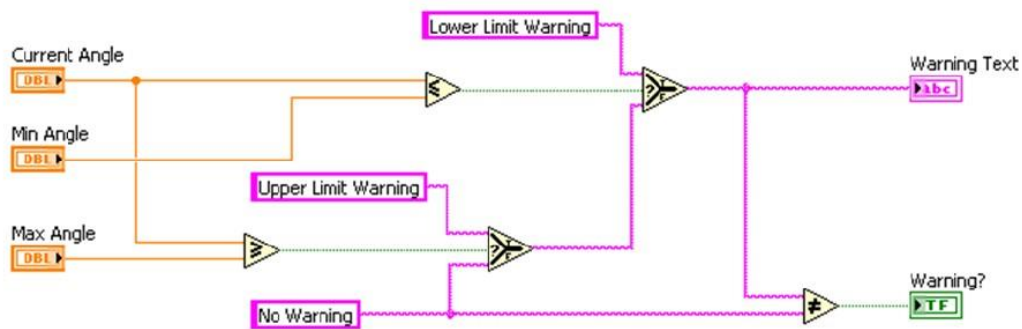


Figure 4.3 Determine Warnings VI Block Diagram

10. Compare Current Angle and Max Angle.



- ❖ Add a Greater Or Equal? function to the block diagram.
- ❖ Wire the **Current Angle** control to the **x** input of the Greater Or Equal? function.
- ❖ Wire the **Max Angle** control to the **y** input of the Greater Or Equal? function.

11. Compare Current Angle and Min Angle.



- ❖ Add a Less Or Equal? function to the block diagram.
- ❖ Wire the **Current Angle** control to the **x** input of the Less Or Equal? function.
- ❖ Wire the **Min Angle** control to the **y** input of the Less Or Equal? function.

12. If the **Current Angle** is equal to or greater than the **Max Angle**, generate a Upper Limit Warning string, otherwise generate a No Warning string.



- ❖ Add the Select function to the block diagram to the right of the Greater Or Equal? function.
- ❖ Wire the output of the Greater Or Equal? function to the **s** input of the Select function.



- ❖ Add a string constant to the block diagram to the upper left of the Select function.
- ❖ Enter Upper Limit Warning in the string constant.
- ❖ Wire the Upper Limit Warning string to the **t** input of the Select function.
- ❖ Hold down the <Ctrl> key and click and drag the Upper Limit Warning string constant to the lower left of the Select function to create a copy of the constant.
- ❖ Enter No Warning in the second string constant.
- ❖ Wire the No Warning string to the **f** input of the Select function.

13. If the **Current Angle** is equal to or less than the **Min Angle**, generate a Lower Limit Warning string, otherwise use the string generated in step 12.

- ❖ Create a copy of the Select function and place it to the right of the Less Or Equal? function.
- ❖ Wire the output of the Less Or Equal? function to the **s** input of the Select function.
- ❖ Create a copy of the string constant and place it to the upper left of the Select function.
- ❖ Enter Lower Limit Warning in the string constant.
- ❖ Wire the Lower Limit Warning string to the **t** input of the Select function.
- ❖ Wire the output of the previous Select function to the **f** input of the new Select function.

14. Display the generated text.

- ❖ Wire the output of the second Select function to the Warning Text indicator.

15. Generate the **Warning?** Boolean control by determining if the value of **Warning Text** is equal to No Warning.



- ❖ Add a Not Equal? function to the left of the **Warning?** Boolean control.
 - ❖ Wire the output of the second Select function to the **x** input of the Not Equal? function.
 - ❖ Wire the No Warning string constant to the **y** input of the Not Equal? function. ❖ Wire the output of the Not Equal? function to the Warning? control.
16. Document the code using the following suggestions on the front panel.
 - ❖ Create tip strips for each control and indicator stating the purpose and units of the object. To access tip strips, right-click an object, and select **Description and Tip**.
 - ❖ Document the VI Properties, giving a general description of the VI, a list of inputs and outputs, your name, and the date the VI was created. To access the **VI Properties** dialog box, select **File»VI Properties**.
 - ❖ Document the block diagram algorithm with a free label.
 17. Save the VI.

■ Test

1. Test the VI by entering a value for **Current Angle**, **Max Angle**, and **Min Angle**, and running the VI for each set.
Table 4.1 shows the expected **Warning Text** string and **Warning?** Boolean value for each set of input values.

Table 4.1 Testing Values for Determine Warnings VI

Current Angle	Max Angle	Min Angle	Warning Text	Warning?
-30	15	-15	Lower Limit Warning	True
0	15	-15	No Warning	False
30	15	-15	Upper Limit Warning	True

What happens if you input a **Max Angle** value that is less than the **Min Angle**? What would you expect to happen?

2. Save the VI.
3. This code runs only one time. Try to use **WHILE** or **FOR** loop in block diagram to run this this code either continuously until you stop it or for specific number of times.

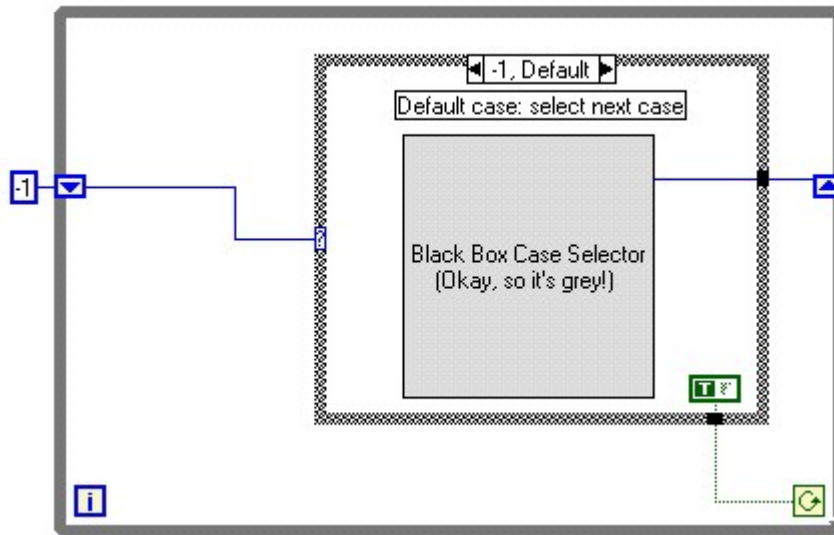
5. Exercise 5: State Machine VI (Create a VI that implements a state machine using a type-defined enum)

- **Scenario:** You must design a template for a user interface state machine. The state machine must also allow for expansion, because processes may be added in the future.

■ Overview

The state machine is one of the fundamental architectures NI LabVIEW developers frequently use to build applications quickly. Developers use state machines in applications where distinguishable states exist. Each state can lead to one or multiple states and can end the process flow. A state machine relies on user input or in-state calculation to determine which state to go to next. Many applications require an “initialize” state followed by a default state, where you can perform many different actions. These actions depend on previous and current inputs as well as states. You can use a “shutdown” state to perform cleanup actions. ■ **Table of Contents**

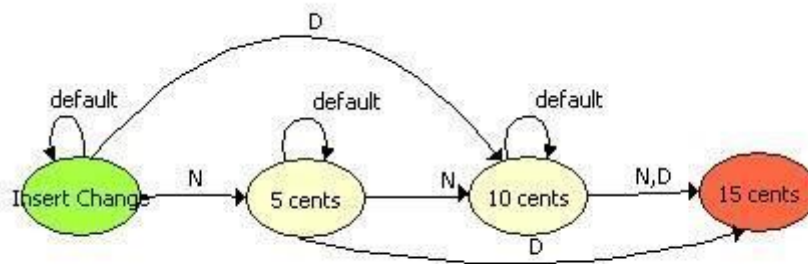
In LabVIEW software, you can create a basic state machine with a while loop, a shift register, a case statement, and some form of case selector (case selectors are discussed in a later section). The while loop is the main program loop, which executes until the conditions for exiting the program are met. The while loop’s main responsibility is to call the case selector and then execute the appropriate case. The shift register keeps track of which case should execute next. Finally, each case of the case statement contains the action for one specific use action. Often the default case is used as the place to check the case selector (in other words, if the user did nothing, check again to see if he has done something yet).



▪ State Diagram

When designing state machines, you can create a state diagram to graphically represent the different states and how they interact. Use state diagrams, the design frameworks for state machines, to model the control algorithms you need with discrete logical states. State Diagrams make it easy to develop and understand the functionality of an application that uses a state machine.

The figure below is an example of a state diagram. The ovals represent the states and the arrows represent the possible transitions between states.



All applications require an initial state, or starting point, followed by transition states that perform different actions. A terminal state, or ending point, is the final state executed and performs cleanup actions. State diagrams are useful in simplifying the design process of applications that use complex decision-making algorithms. To create an effective state diagram, you must know the various states of the application and how they relate to one another. By visualizing the various execution states of the application, you improve the overall design of the application.

▪ Design a State Machine with a State Diagram

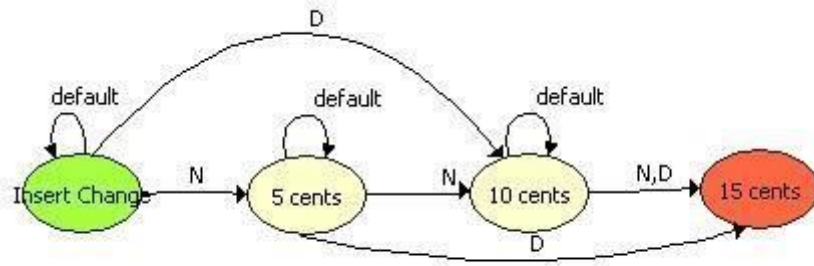
Imagine a vending machine that accepts combinations of nickels and dimes to get a coke. The cost of a coke is 15 cents and the machine does not return change. First, establish the states that the vending machine might be in:

- Start: No money inserted
- 5 cents
- 10 cents
- Done: 15 cents

Now think about the possible ways or paths that the vending machine can take to get into these states. Don't forget to consider how the vending machine starts in the initial state.

- Nickel is inserted
- Dime is inserted
- Default: Nothing is inserted

From these lists, you see that there are four states and three possible paths from each state. You need to depict which states are connected by which paths. For example, when the vending machine is in the initial start state, the total change inserted is 0 cents. When a nickel is inserted, the vending machine must go to the 5 cent state. Therefore, the start state leads to the 5 cent state by the nickel path. By considering all states and paths, you can create a state diagram for the vending machine:

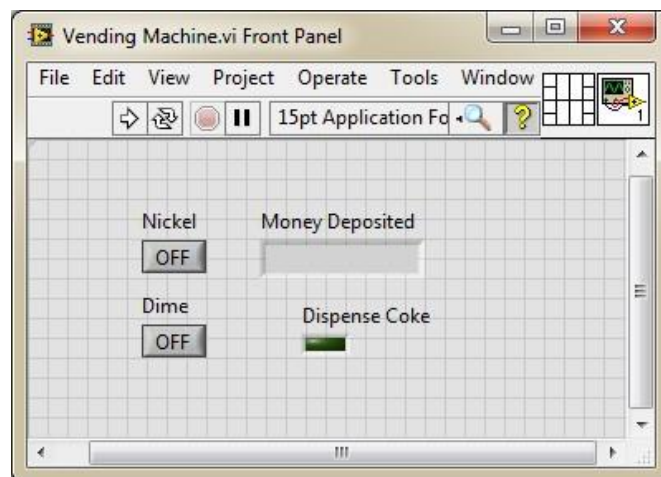


With a state diagram, you can better understand how to create a state machine. ▀

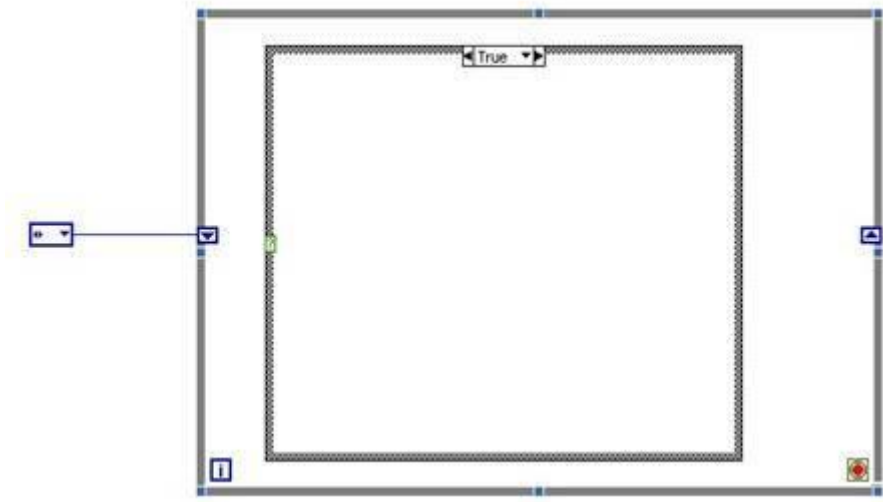
Building a State Machine

Using the state diagram above, create a state machine.

1. Create a new blank VI.
2. On the front panel place:
 - Two text buttons named “Nickel” and “Dime”
 - Text indicator named “Money Deposited”
 - Boolean indicator named “Dispense coke”

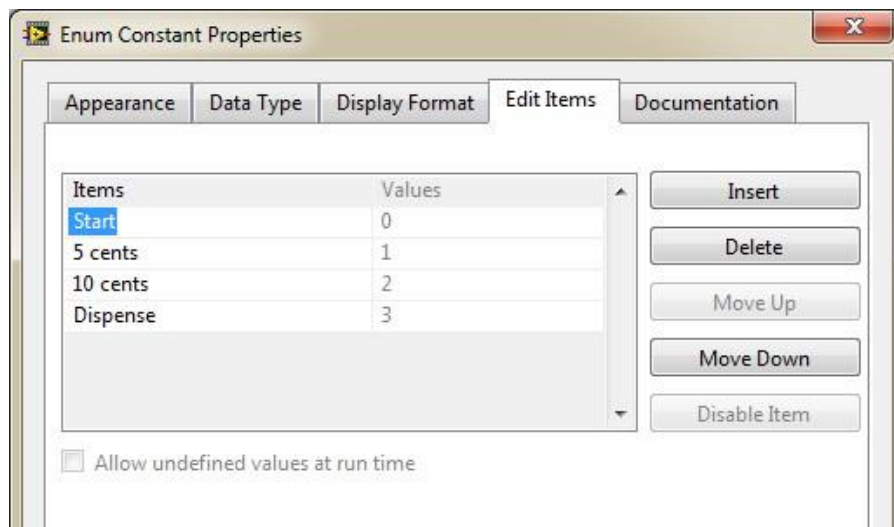


3. Place a while loop on the block diagram.
4. Place a case structure in the while loop.
5. Create a shift register on the while loop.
6. Create an Enum and wire it to the shift register to initialize it.

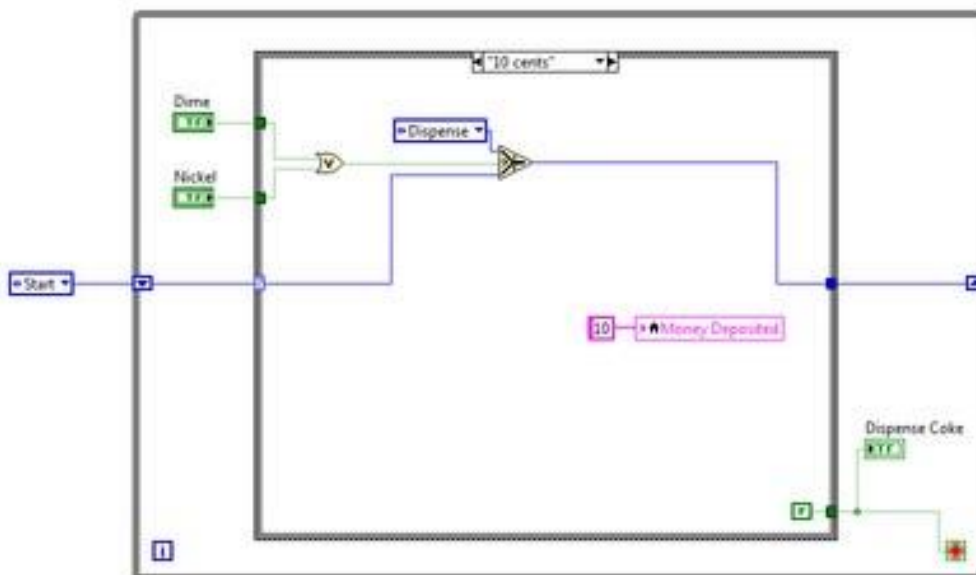
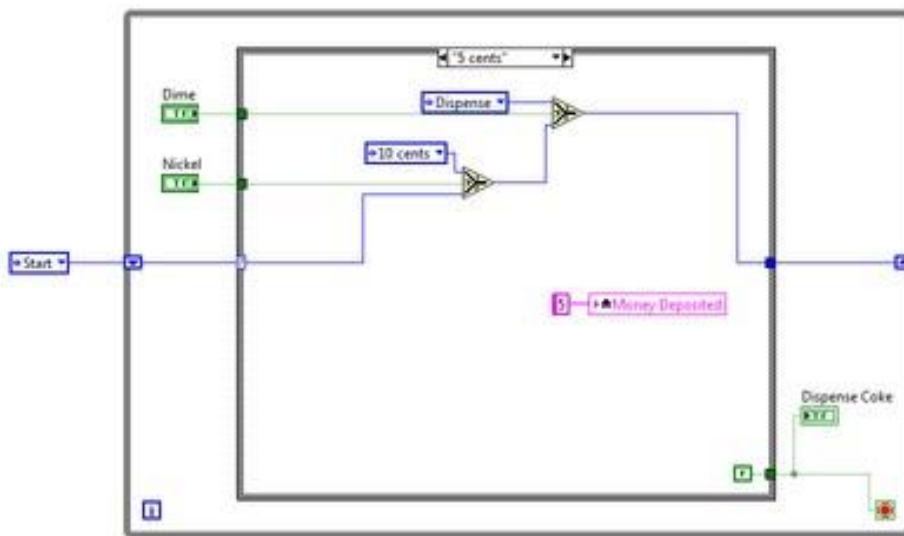
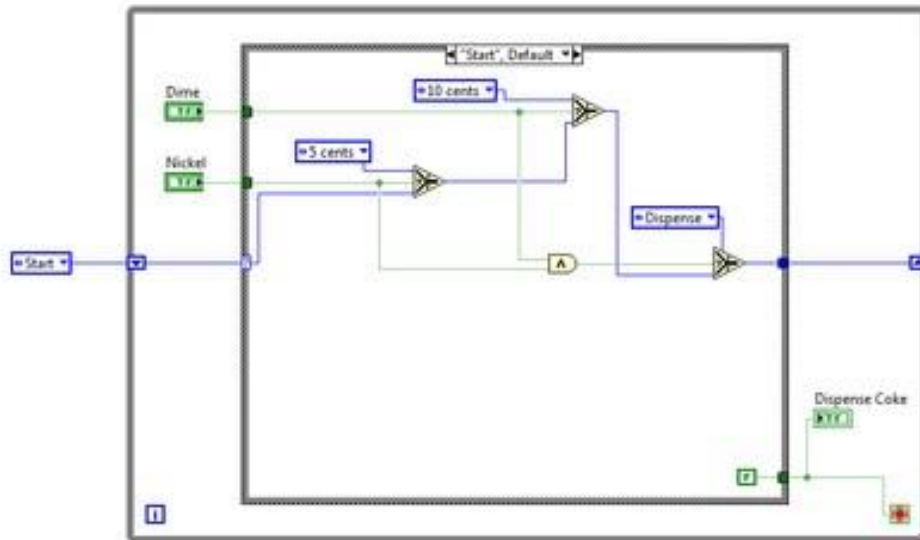


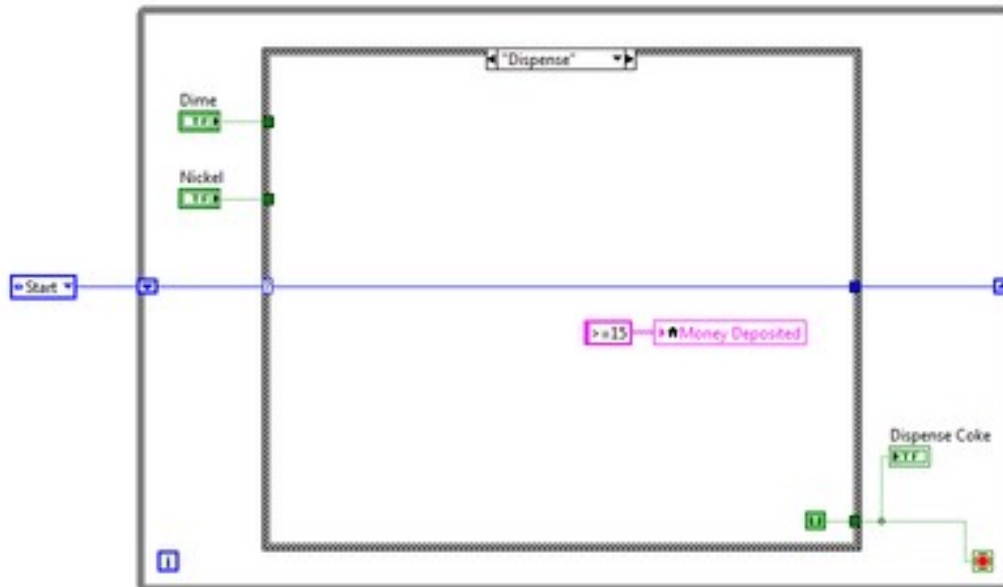
7. Right-click the Enum, select Edit Items, and add the following “states”:

- Start
- 5 cents
- 10 cents
- Dispense



8. Wire the shift register to the conditional input of the case structure.
9. Right-click the case box at the top of the case structure and select Add Case for Every Value.
10. Wire the different cases as depicted in the following figures.





11. Wire Dispense coke to the Boolean output of the case structure.
 12. Inside the while loop, place a wait function with a constant.
 13. Outside of the while loop, wire a constant of 0 to the Money Deposited string indicator.
- The finished VI should look like this:

